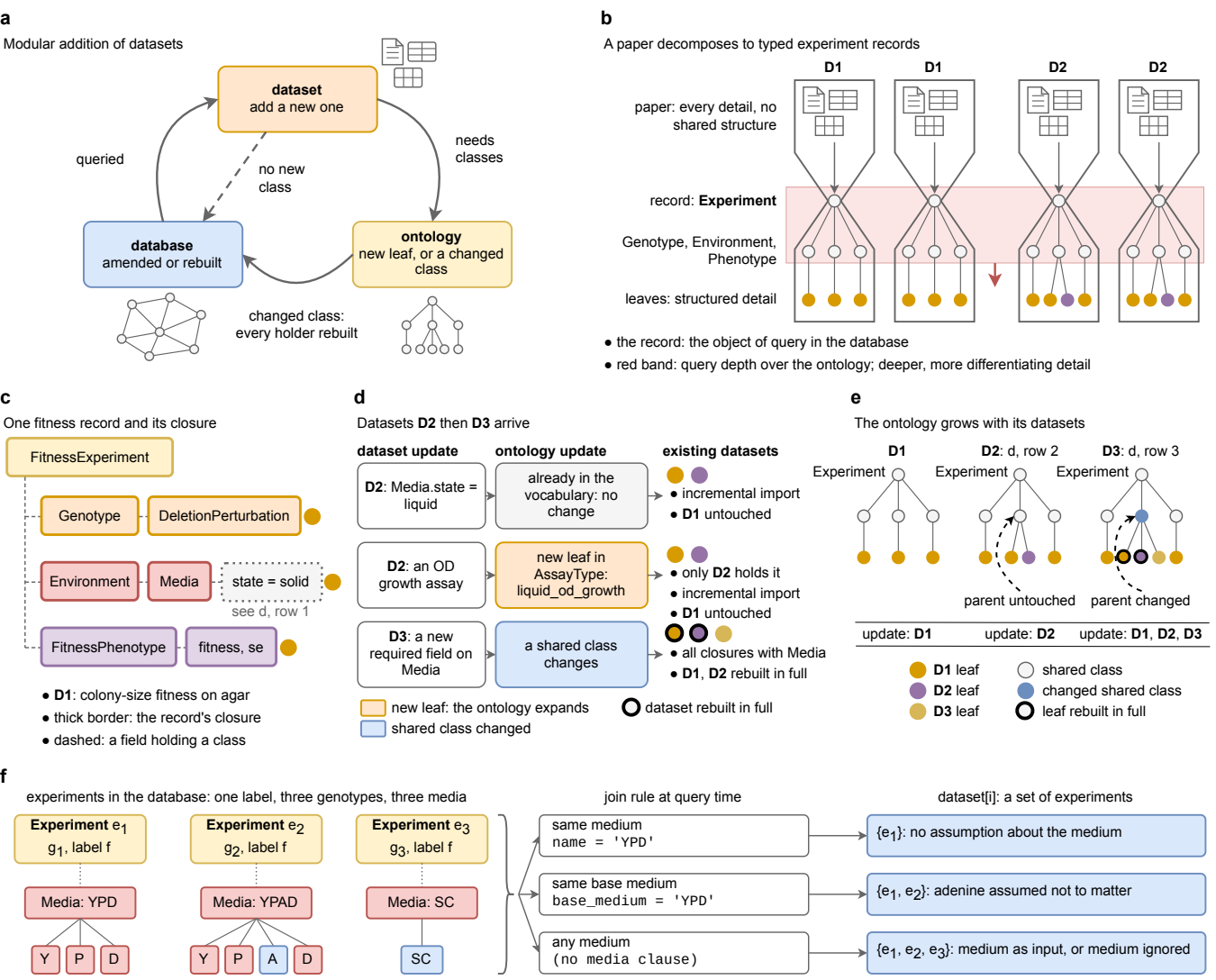

a
Modular addition of datasets
dataset
add a new one
needs
classes
ontology
new leaf, or a changed
class
database
amended or rebuilt
queried
no new
class
changed class:
every holder rebuilt
b
A paper decomposes to typed experiment records
D1
D1
D2
D2
paper: every detail, no
shared structure
record: Experiment
Genotype, Environment,
Phenotype
leaves: structured detail
• the record: the object of query in the database
• red band: query depth over the ontology; deeper, more differentiating detail
c
One fitness record and its closure
FitnessExperiment
Genotype
DeletionPerturbation
Environment
Media
state = solid
see d, row 1
FitnessPhenotype
fitness, se
• D1: colony-size fitness on agar
• thick border: the record's closure
• dashed: a field holding a class
d
Datasets D2 then D3 arrive
dataset update
ontology update
existing datasets
D2: Media.state =
liquid
already in the
vocabulary: no
change
• incremental import
• D1 untouched
D2: an OD
growth assay
new leaf in
AssayType:
liquid_od_growth
• only D2 holds it
• incremental import
• D1 untouched
D3: a new
required field on
Media
a shared class
changes
• all closures with Media
• D1, D2 rebuilt in full
new leaf: the ontology expands
shared class changed
dataset rebuilt in full
e
The ontology grows with its datasets
D1
D2: d, row 2
D3: d, row 3
Experiment
Experiment
Experiment
parent untouched
parent changed
update: D1
update: D2
update: D1, D2, D3
D1 leaf
D2 leaf
D3 leaf
shared class
changed shared class
leaf rebuilt in full
f
experiments in the database: one label, three genotypes, three media
Experiment e1
g1, label f
Experiment e2
g2, label f
Experiment e3
g3, label f
Media: YPD
Media: YPAD
Media: SC
Y
P
D
Y
P
A
D
SC
join rule at query time
same medium
name = 'YPD'
same base medium
base_medium = 'YPD'
any medium
(no media clause)
dataset[i]: a set of experiments
{e1}: no assumption about the medium
{e1, e2}: adenine assumed not to matter
{e1, e2, e3}: medium as input, or medium ignored